PARTNERSHIP GUIDE

between

THE UNIVERSITY OF MANCHESTER
Systems and Software Security (S3) Research Group
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Guide Reference	PG-S3-v01
Date of Issue	01 May 2026
Language	English (authoritative); additional translations may be appended as informative annexes.

Preamble

This Partnership Guide (hereinafter "Guide") is entered into by and between the University of Manchester, through its Department of Computer Science and the Systems and Software Security (S3) Research Group, and the organization identified on the cover page as the Partner (hereinafter "Partner" or "Commercial Partner"), collectively referred to as the "Parties".

A formal Technical Cooperation Agreement (TCA) will be drafted and signed between the S3 Group and the Partner whenever a joint project is submitted to a UK research funding body or other applicable funder. That TCA will reference this Guide and incorporate the terms relevant to the specific grant. Until then, this Guide serves as the shared reference framework for the partnership.

1. Definitions

For this Guide, the following terms shall have the meanings ascribed to them below:

- "Lead Research Organization" or "LRO" means the higher education or research institution identified on the cover page of this Guide, including its Systems and Software Security (S3) Research Group.
- "S3 Group" means the Systems and Software Security Research Group within the Lead Research Organization, as identified on the cover page.
- "Partner" means the commercial, industrial, or research organization identified on the cover page of this Guide.
- "Project" means any specific collaborative initiative agreed upon by the Parties under the framework established by this Guide.
- "Funding Body" means any public or private research funding agency, foundation, council, or governmental body that provides financial support for research projects, including but not limited to national research councils, innovation agencies, and international funding programs.
- "Project Partner" means an organization that contributes directly (cash) or indirectly (in-kind) to a funded project without directly receiving grant funding from the Funding Body.
- "Visiting Researcher" means a researcher nominated by the Partner who carries out work at the Lead Research Organization for a limited period under this Guide.
- "Intellectual Property" or "IP" means all inventions, discoveries, patents, trademarks, copyrights, trade secrets, know-how, and any other proprietary rights.
- "Confidential Information" means any non-public technical, commercial, financial, or research information exchanged between the Parties.

2. Objective and Purpose

The purpose of this Guide is to establish a formal framework for technical and scientific cooperation between the Lead Research Organization and the Partner, enabling:

- Joint research and development activities in software verification, formal methods, cybersecurity, and AI-assisted systems;

- Mutual access to expertise, tools, infrastructure, and international networks;
- Submission of joint funding applications to national and international research funding bodies, innovation agencies, and charitable foundations;
- Mobility and exchange of researchers and technical specialists;
- Co-development of innovative solutions, methodologies, and potential commercial products;
- Co-authorship of academic publications, conference papers, and technical reports.

3. Scope of Cooperation

The technical scope of this Guide encompasses the following areas, which may be expanded by mutual written consent:

3.1 Primary Technical Areas

- Software Verification: model checking, bounded model checking, static analysis, and theorem proving;
- Formal Methods applied to embedded systems, cyber-physical systems, IoT, and safety-critical applications;
- AI-assisted verification: integration of Large Language Models (LLMs) with formal verifiers;
- Cybersecurity: vulnerability detection, cryptographic protocols, blockchain smart contract verification;
- Hardware verification: ARM architecture (Morello/CHERI), SoC, and multi-core processor formal analysis.

3.2 Specific Activities

- Provision of letters of support for joint grant applications;
- Definition and co-development of industrial use cases relevant to the Partner's technical challenges;
- Access to and collaborative use of the Lead Research Organization's research tools and platforms;
- Organization of joint workshops, seminars, and technical training sessions;
- Secondment of Partner researchers as Visiting Researchers at the Lead Research Organization;
- Preparation of joint technical reports and deliverables for funded projects.

4. Mutual Benefits and Value Proposition

The Parties recognize that this cooperation creates reciprocal and complementary value. The following benefits are not exhaustive and may be supplemented by specific Project Annexes.

Benefits to the Partner	Benefits to the Lead Research Organization
Early access to cutting-edge software verification and AI tools developed by the S3 Group.	Access to real-world industrial use cases from the Partner's sector and region, strengthening impact evidence for funding applications.
Strengthening the Partner's R&D capability with high cost-effectiveness, leveraging world-class academic expertise.	Co-development of solutions applicable to novel technical contexts, potentially generating joint patents and IP.
Direct connection to the S3 Group's global innovation ecosystem, spanning industry, government, and academia.	Expansion of the international partner network into the Partner's market and region.
Possibility of co-funded or fully funded collaborative projects through national and international research funding bodies.	Strengthening the Research Organization with concrete industrial outcomes and stakeholder evidence.
Joint academic publications, international visibility, and co-authorship in top-tier conferences (CAV, ICSE, TACAS).	Attraction of qualified research talent through the Visiting Researcher program.
Integration of new software security and formal verification solutions directly applicable to the Partner's products and services.	International dissemination of research results, supported by a commercially relevant and credible partner base.

5. Partner Roles and Types of Engagement

The Parties agree that individuals nominated by either Party to participate in joint Projects shall be assigned formal roles in accordance with the classification framework set out in each call.

Note on Visiting Researchers:

Partner nominees who carry out research at the Lead Research Organization for a defined period under this Guide will be registered as Visiting Researchers (or equivalent institutional category). The Lead Research Organization does not employ them and must comply with all applicable visa, immigration, and security clearance requirements of the host country. The Lead Research Organization's designated technical contact will provide full onboarding support.

6. Types of Support Provided by the Lead Research Organization

In the context of this Guide, the Lead Research Organization and the S3 Group commit to providing the following categories of support to the Partner, subject to capacity and the terms of each Project:

6.1 Scientific and Technical Support

- Access to the ESBMC verification platform, associated tools, documentation, and training;
- Technical consultation on software verification, formal methods, LLM-verifier integration, and cybersecurity;
- Joint definition of research problems and co-design of solution approaches;
- Co-supervision of technical work carried out by Partner researchers or Visiting Researchers.

6.2 Grant and Funding Support

- Preparation and co-authorship of grant applications to national and international research funding bodies and innovation agencies;
- Inclusion of the Partner as a named Project Partner in grant submissions, with appropriate in-kind or direct contributions;
- Provision of letters of support documenting the Partner's role, relevance, and anticipated benefits;
- Guidance on compliance with applicable funder policies, open access requirements, and impact reporting standards.

6.3 Mobility and Exchange Support

- Issuance of formal invitation letters and support documentation for Visiting Researcher visa applications (Skilled Worker / Standard Visitor);
- Assistance with ATAS (Academic Technology Approval Scheme) certificate applications, where required;
- Registration of Partner researchers as Visiting Researchers or PGR Visiting Students at the Lead Research Organization;
- Onboarding support: access to the Lead Research Organization systems, library, laboratories, computing infrastructure, and institutional email.

6.4 Dissemination and Publication Support

- Co-authorship of academic papers submitted to peer-reviewed journals and top-tier conferences;
- Review of technical reports, deliverables, and patent applications jointly developed;
- Facilitation of participation in international events, workshops, and conferences.

7. Partner Definition, Commitments, and Contribution Types

Definition of Project Partner

"Project Partner" means an organization that contributes to a funded Project through direct (cash) and/or indirect (in-kind) support. The funding status of a Project Partner (whether grant-receiving or not) shall be determined by the applicable Funding Body and specified in the relevant Project Annex.

7.1 General Commitments

Partners engaging through this Guide are expected to:

- Designate a named technical point of contact responsible for coordinating all activities under this Guide;
- Provide a specific industrial use case or technical challenge to be addressed;
- Issue letters of support when requested for grant applications where the Partner's involvement is material;
- Participate in scheduled technical meetings with the S3 Group team;
- Nominate qualified individuals for Visiting Researcher roles and ensuring compliance with applicable immigration, visa, and security clearance requirements of the host country;
- Maintain confidentiality of shared information in accordance with Clause 9;
- Notify the Lead Research Organization of any material change in organizational circumstances that may affect the collaboration.

7.2 Types of Contribution

Partner contributions are formally classified into two categories for the purposes of grant applications and project reporting. The Partner shall declare, at the time of each Project Annex, which type(s) of contribution it will provide and the corresponding estimated value.

Type	Description & Examples	Valuation
DIRECT *(Cash / tangible resource)*	Financial or tangible contributions are transferred to or placed at the direct disposal of the project. Examples include: • Cash funding or financial sponsorship of project activities; • Donated or loaned equipment (hardware, embedded systems boards, test rigs); • Licensed software tools or commercial simulation platforms made available exclusively for the project; • Staff formally seconded to the project team (charged at cost); • Payment of travel, accommodation, or subsistence costs for joint activities.	Declared as a monetary amount (£) in the grant application.

Type	Description & Examples	Valuation
INDIRECT *(In-kind / non-cash)*	Non-financial contributions that represent real value to the project without a cash transfer. Examples include: • Access to proprietary datasets, industrial benchmarks, or test environments; • Use of Partner's technical infrastructure, laboratory facilities, or computing resources; • Expert engineering or technical hours dedicated to joint activities (e.g., co-design meetings, code reviews, requirements elicitation); • Provision of real-world case studies, product documentation, or system specifications; • Internal software licenses, SDKs, or APIs made available to the research team; • Participation in workshops, validation sessions, and dissemination events.	Declared as a monetary equivalent (£) if quantifiable, or as £0 if the contribution has no direct financial value.

7.3 Contribution Declaration per Project

For each Project Annex executed under this Guide, the Partner shall declare:

- Whether the contribution is Direct, Indirect, or both;
- The estimated monetary value (in GBP) of each contribution type; if an indirect contribution has no quantifiable monetary value, it shall be declared as £0;
- The name(s) and role(s) of individuals from the Partner organization who will carry out or supervise the contribution.

Important:
The role of the Partner under this Guide is, in principle, that of a non-grant-receiving Project Partner, contributing to the Project through direct and/or indirect (in-kind) support. However, this classification may vary depending on the rules and structure of the applicable Funding Body. In certain funding schemes, the Partner may be eligible to receive grant funding directly or indirectly. In such cases, the terms governing funding allocation, eligibility, and financial flows shall be determined in accordance with the requirements of the Funding Body and set out in the relevant Project Annex or formal agreement. Unless otherwise specified in a Project Annex or required by the applicable Funding Body, the Partner's contributions are expected to flow from the Partner to the Project rather than from the Project to the Partner.

8. Periodic Review

This Guide has no fixed expiry date. It remains valid until superseded by a revised version or until the partnership is formally concluded by mutual written notice.

A review of the partnership's activities and outcomes is encouraged at least annually, or at any time either party requests it. Reviews may be conducted informally and documented in a brief shared note.

9. Intellectual Property, Patents, Confidentiality, and Data Protection

Lead Research Organization as a Charity - Partner Freedom of Choice

The Lead Research Organization is a not-for-profit institution whose primary purpose is education and the advancement of knowledge for public benefit. This institutional purpose has a direct practical consequence for commercial partners: the Lead Research Organization does not impose a fixed model for IP ownership, licensing, or commercialization. The Parties are free to negotiate and agree on the arrangement that best serves their mutual interests and the public-benefit objectives of the research, on a project-by-project basis. All such terms may vary depending on the requirements of the applicable Funding Body.

9.1 Intellectual Property - General Framework

The Parties acknowledge that intellectual property arrangements under this Guide are intended to be flexible and determined on a project-by-project basis, taking into account the requirements of the applicable Funding Body, the nature of the collaboration, and the mutual interests of the Parties.

The provisions set out below are non-binding default principles provided solely as a reference framework and starting point for discussion. They do not create any presumption or obligation and may be modified, replaced, or disapplied in whole or in part by agreement in a Project Annex or in a formal Technical Cooperation Agreement (TCA), which shall take precedence. In the absence of a project-specific agreement, the Parties may refer to the following indicative principles:

Category	Default Position (may be varied by Project Annex)
Background IP	Each Party retains full, sole ownership of all intellectual property created or developed independently of this Guide, whether before or during its term. Neither Party acquires any rights over the other's Background IP except as expressly licensed in a Project Annex.
Foreground IP (sole)	IP created solely by one Party in the course of a joint Project remains owned by that Party. The other Party may be granted a license on terms agreed in the Project Annex.
Foreground IP (joint)	IP created jointly by both Parties shall be co-owned in proportion to each Party's contribution, as assessed and agreed at the time of the relevant Project Annex. Joint owners each retain the right to exploit the IP without the other's consent, unless otherwise restricted in writing.
LRO's institutional purpose	The Lead Research Organization's primary IP interest is in enabling public benefit through open publication and dissemination. The Partner is free to propose exclusive or non-exclusive licensing arrangements, assignment of rights, or co-ownership models. The LRO will assess each proposal against its institutional objectives and applicable law.
Open-source outputs	Research tools, code, datasets, and benchmarks arising from funded Projects are typically released under open-source licenses (e.g., Apache 2.0, MIT) unless a Project Annex specifies otherwise, in accordance with the applicable funder's open-access and open-research-data policies.
Commercial arrangements	Licensing, assignment, or other commercialization models shall be determined on a project-specific basis.

9.2 Patents

Patent strategy is agreed on a project-by-project basis and will be defined in the formal Technical Cooperation Agreement for each specific grant submission. The guidelines below represent general good practice and will inform, but not replace, the project-specific patent agreement. Where a joint activity results in a potentially patentable invention:

- Either Party may notify the other in writing of a potential invention.
- The disclosing Party shall not publish or publicly disclose the invention, to preserve patentability.
- If both Parties wish to pursue a patent jointly, costs, ownership share, territorial coverage, and licensing rights shall be agreed in a separate Patent Agreement or recorded in the relevant Project Annex.
- If only one Party wishes to pursue a patent, it may do so at its own cost, subject to a fair license being granted to the other Party on request, on terms to be agreed.
- The Lead Research Organization's Technology Transfer Office (or equivalent unit) will be involved in any patent filing by the LRO or on the LRO's behalf. The Partner is encouraged to involve its own IP counsel.
- The Partner is free to seek sole patent ownership for inventions arising primarily from its own resources and expertise, provided the Lead Research Organization's contribution is acknowledged and appropriately licensed.

9.3 Non-Disclosure Agreement (NDA) and Confidentiality

The obligations set out in this clause establish the default confidentiality framework governing the cooperation between the Parties. A separate non-disclosure agreement (NDA) is not generally required; however, the Parties may agree to enter into a specific NDA where appropriate, including for particular Projects, disclosures, or funding body requirements.

Each Party agrees to treat all Confidential Information received from the other Party with the same standard of care it applies to its own confidential information, and in no case less than a reasonable standard of care.

Confidential Information may be disclosed only to employees, contractors, or collaborators of the receiving Party on a strict need-to-know basis, provided that such persons are bound by confidentiality obligations no less restrictive than those set out in this Guide.

Confidential Information shall be used solely for the purposes of the partnership and the joint Projects contemplated under this Guide.

The following shall not be considered Confidential Information:

- Information that is or becomes publicly available through no fault of the receiving Party;
- Information that was already in the receiving Party's possession at the time of disclosure, free of any confidentiality obligation;

- Information independently developed by the receiving Party without use of or reference to the Confidential Information;
- Information required to be disclosed by applicable law, regulation, or court order, provided that the disclosing Party is notified promptly and in advance where legally permitted.

Academic publication and the Lead Research Organization's charitable mission:

Researchers at the Lead Research Organization retain the right to publish research results for academic and public benefit purposes. Before any publication containing potentially confidential or patentable material, the Partner shall receive a written draft with a minimum 30-day review period. During this period, the Partner may request redactions of genuine trade secrets or a delay in filing a patent. The Lead Research Organization cannot agree to suppress publication indefinitely, consistent with its institutional objectives and applicable funder's open-access obligations.

9.4 Data Protection and GDPR

Both Parties shall comply with all applicable data protection and privacy legislation in their respective jurisdictions. This includes, but is not limited to, the General Data Protection Regulation (GDPR) and its national implementing legislation, the UK GDPR, the Brazilian Lei Geral de Proteção de Dados (LGPD), or any equivalent data protection framework applicable to the Parties.

- Personal data shall only be processed for specified, explicit, and legitimate purposes. Processing must be lawful, fair, and transparent.
- Where personal data is shared between the Parties, both Parties shall agree on their respective roles as Controller, Processor, or Joint Controller before any transfer occurs, and shall document this in a Data Sharing Agreement or the relevant Project Annex.
- Research activities under this Guide may benefit from research exemptions provided under applicable data protection legislation, which allow personal data to be processed and retained for research purposes beyond the original collection purpose, provided appropriate safeguards are in place.
- The Parties shall implement appropriate technical and organizational measures (TOMs) to protect personal data against unauthorized access, loss, or destruction - including pseudonymization, encryption, and access controls where appropriate.
- Data Protection Impact Assessments (DPIAs) - or equivalent risk assessments under applicable law - shall be conducted where required, particularly for processing likely to result in high risk to the rights and freedoms of individuals.
- The Lead Research Organization's Data Protection Officer (DPO) or equivalent designated officer shall be the primary point of contact for data protection queries on the LRO's side. The Partner is responsible for its own compliance and is advised to consult its own DPO or legal counsel.

Cross-border data transfers: Where personal data is transferred between countries, the Parties shall ensure that an appropriate transfer mechanism is in place - such as Standard Contractual Clauses,

an adequacy decision, binding corporate rules, or an equivalent instrument recognized under applicable law - prior to any such transfer.

Key reference on data protection in research:
Both Parties are encouraged to consult the data protection guidance published by their respective national supervisory authorities. As a widely referenced framework, the GDPR and Research overview published by UKRI/ICO (ukri.org) and the ICO Research guidance at ico.org.uk provide practical principles applicable across international research collaborations.

9.5 Partner's Freedom to Negotiate

The Lead Research Organization acknowledges that commercial partners operate in different regulatory, fiscal, and strategic environments worldwide. Accordingly:

- The Partner is free to propose any IP, licensing, NDA, or data governance arrangement it considers appropriate for each Project, including exclusive licenses, assignment of rights, tiered royalty structures, or first-refusal options.
- The Lead Research Organization will engage with all proposals in good faith, assess them against its institutional objectives and applicable funder requirements.
- The Parties may engage their respective legal, IP, and technology transfer teams at any stage to negotiate specific arrangements without prejudice to this Guide.
- Nothing in this Guide prevents the Partner from entering parallel or successor commercial agreements with the Lead Research Organization's technology transfer function for the purpose of licensing, spin-out creation, or product commercialization.

10. Financial Arrangements

The Partner's contribution under this Guide is principally in-kind, comprising: expert time, provision of industrial use cases, access to internal data and infrastructure, participation in joint activities, and support for grant applications.

Where a joint grant application is successful, the allocation of funds shall be governed by the terms of the relevant funding body and documented in a separate sub-agreement or Project Annex.

11. Institutional Contacts

LRO - Academic Lead	**Prof. Dr. Lucas Carvalho Cordeiro** Professor in the Department of Computer Science at the University of Manchester (UoM) & Head of the Systems and Software Security (S3) Research Group. lucas.cordeiro@manchester.ac.uk
LRO - Commercial Contact	**Dr. Pierre Vilar Dantas** Postdoctoral Researcher & Commercial Lead pierre.dantas@manchester.ac.uk

Chapter 12

Commercialisation Hypothesis

Framework, Obligations, and Guidance for Partners

12.1 Purpose and Context

A Commercialization Hypothesis is a structured, forward-looking statement that articulates how a research project, if successful, could eventually move out of the laboratory and into a deployed technology, product, or business. It is not a business plan, nor a binding commitment to commercialize - it is a baseline document that captures the team's best current thinking about the downstream potential of their work.

Why it matters for this partnership:

For Projects under this Guide that involve or target funding from bodies requiring a formalized commercialization plan, a Commercialization Hypothesis is a mandatory submission alongside the full project application. Even when not formally required, articulating a clear commercialization pathway strengthens the case for industrial relevance - an area where the Partner's participation is crucial evidence. The commercial partner's role in co-shaping the hypothesis is itself a signal of real-world demand.

12.2 Commercialization Hypothesis Framework - What is Required

Many leading research funding bodies require project creators to submit a Commercialization Hypothesis at the start of their project. The rationale is consistent across funders: research impact is achieved only when the work leads to technologies and products deployed at scale, benefiting society and the economy broadly.

The hypothesis is not typically a primary criterion for award selection - it is a baseline document that helps funders understand the team's thinking and identify downstream opportunities. The hypothesis is a starting point for dialogue, not a scorecard.

The Commercialization Hypothesis should address the following five questions. The format requirements (length, font, file type) will be specified by the applicable Funding Body for each submission:

#	Commercialization Hypothesis Question
1	What is your best guess of how your proposed research, if successful, will eventually make its way out of the lab and into a deployed technology, product, or business?
2	Imagine your project concludes successfully. What does the next stage of development look like? What resources will be needed? Who will need to be involved? How do you expect it will be funded?

#	Commercialization Hypothesis Question
3	Are there members of your core team whose primary interest is to advance this research toward a viable product - whether through licensing, spinout, open source, or otherwise? Are any individuals well-suited to lead such efforts based on interest or experience? (Funded personnel only)
4	Rate your core team's experience and comfort (1–10) with: IP strategy; market landscape and needs; competitive landscape; applications and product definition; fundraising from private investors; techno-economic analysis; supply chain and manufacturing; social, ethical, and regulatory frameworks. Could the funding increase your team's understanding of these areas? If so, how? (Funded personnel only; honesty encouraged - no advantage to higher ratings)
5	Who will manage decisions related to intellectual property for the project? What activities or agreements do you expect will be needed to protect the IP developed and maximize the probability and scale of impact?

12.3 Technology Readiness Levels (TRL)

Technology Readiness Levels (TRL) are an internationally recognized measurement system used to assess the maturity of a technology at any given stage of development. Originally developed by NASA and now adopted by research councils, innovation agencies, the European Commission, the US Department of Defense, and other major funding bodies worldwide, TRLs provide a common language for researchers, industrialists, and funders to assess where a technology sits on the path from fundamental concept to operational deployment.

A technology project is evaluated against defined parameters for each level and assigned a TRL rating based on its progress. Funding bodies use TRLs to determine whether a project or proposal is suitable for a specific funding opportunity - guidance for each opportunity states which TRL range is eligible.

TRL	Label	Definition
TRL 1	**Basic principles**	Basic principles observed and reported. Scientific research begins to be translated into applied research. No experimental proof yet.
TRL 2	**Concept formulated**	Technology concept or application formulated. Practical applications can be identified, though evidence and detail are still limited.
TRL 3	**Proof of concept**	Analytical and experimental critical function or characteristic proof-of-concept. Active research and development commence. Studies and laboratory measurements validate analytical predictions.
TRL 4	**Lab validation**	Technology basic validation in a laboratory environment. The basic technological components are integrated to ensure they work together.
TRL 5	**Relevant environment validation**	Technology basic validation in a relevant environment. The basic technological components are integrated with reasonably realistic supporting elements so that the technology can be tested in a simulated environment.

TRL	Label	Definition
TRL 6	**Prototype demonstration**	Technology model or prototype demonstration in a relevant environment. A representative model or prototype is tested in a relevant environment.
TRL 7	**Operational prototype**	Technology prototype demonstration in an operational environment. A prototype near or at the planned operational system. Testing in an operational environment.
TRL 8	**Technology complete**	Actual technology completed and qualified through test and demonstration. Technology has proven to work in its final form and under expected conditions.
TRL 9	**Mission-proven**	Actual technology qualified through successful mission operations. Technology applied in its final form and under mission conditions.

TRL 1–3 - Research Phase	TRL 4–6 - Development Phase	TRL 7–9 - Deployment Phase
Fundamental science through proof-of-concept. Typical entry point for fundamental research grants from national research councils and discovery-focused funding bodies.	Laboratory to prototype. Typical eligibility for applied research and partnership programs from innovation agencies and industry-facing research councils.	Operational prototype to mission-proven. Typical for knowledge transfer programs, scale-up grants, and commercial licensing or spin-out routes.

12.4 TRL - Indicative Project Positioning

For each Project Annex executed under this Guide, both Parties shall declare the current TRL of the technology and agree on a target TRL to be achieved by the end of the funded period. The following table illustrates a typical TRL trajectory for software verification research, which can be adapted to the specific technology and Project in scope:

Horizon	Illustrative TRL Trajectory and Partner Role
Start of Project (TRL 1–4)	Core theory published and validated analytically or experimentally. Initial tool or framework integration is underway. Industrial partners providing early requirements and validation scenarios.
End of Project (TRL 5–6)	Full framework or prototype integrated into the research platform and validated on real industrial case studies provided by partners. Technology demonstrated in a relevant environment.
Post-Project (TRL 7–8)	The technology prototype demonstrated operationally. Spin-off ventures and/or industrial partners may adopt the technology as a commercial product feature. Licensing or spin-out route pursued via the LRO's technology transfer function.
Long-term (TRL 9)	Technology is deployed in production toolchains, operational environments, or embedded in industry certification and qualification workflows for safety-critical, regulated, or high-value sectors.

12.5 Obligations Under this Guide

For each Project Annex executed under this Guide that targets a Funding Body requiring a Commercialization Hypothesis or Technology Readiness Level (TRL) assessment, the Parties shall comply with the applicable requirements of that Funding Body.

Where such requirements apply, both Parties commit to the following:

- The Project Lead shall prepare a draft Commercialization Hypothesis addressing the relevant questions specified by the applicable Funding Body and share it with the Partner for review;
- The Partner shall contribute to the hypothesis by: (a) providing an informed assessment of market needs and competitive context relevant to its sector; (b) identifying whether individuals within its organization have an interest in advancing the technology toward a product; and (c) contributing to any required self-assessment of the Project's maturity, including TRL where applicable;
- The Parties shall, where required, declare the current TRL of the technology at the start of the Project and agree on a target TRL to be achieved by the end of the funded period;
- The Partner shall disclose any existing commercial agreements, licenses, or IP encumbrances that could reasonably affect the proposed commercialization pathway, where such disclosure is required for the preparation of the hypothesis;
- Where a Funding Body initiates a further commercialization or award negotiation stage, the Parties shall collaborate in good faith to refine the hypothesis and align with the applicable requirements.

12.6 IP Strategy in the Commercialization Hypothesis

Question 5 of the Commercialization Hypothesis specifically asks who will manage IP decisions and what agreements will be needed. The Parties agree that, for each Project:

- IP strategy shall be co-designed by the Lead Research Organization's Technology Transfer function and the Partner's designated IP or legal representative, and documented in the Project Annex.
- The choice between open-source release, patent filing, exclusive licensing, non-exclusive licensing, or assignment shall be driven by the TRL stage, the nature of the innovation, and the commercialization pathway most likely to achieve large-scale deployment.
- At TRL 1–3 (research phase), outputs are typically published as open research in accordance with applicable funder open access requirements and the LRO's open research policy;
- At TRL 4–6 (development phase), a patent filing decision window shall be opened before any public disclosure, as described in Clause 9.2;
- At TRL 7–9 (deployment phase), licensing or spin-out negotiations shall be led by the Lead Research Organization's Technology Transfer function in partnership with the relevant commercial partner, with the Partner having a right of first negotiation for an exclusive license in its designated sector or geography.

APPENDICES

Regulatory, Compliance, and Policy Framework

Annex A - Project Annexes Template

Each specific collaboration under this Guide shall be documented in a Project Annex signed by both Parties. Each Annex shall include at a minimum:

- Project title, objectives, and research questions;
- Timeline, milestones, and deliverables;
- Roles and nominated individuals from both Parties (per the role framework in Clause 5);
- Resource and financial contributions of each Party (per Clause 7.2);
- IP ownership, licensing, and publication arrangements (per Clause 9);
- Data management plan and GDPR roles (Controller / Processor / Joint Controller);
- Expected outcomes, impact metrics, and reporting obligations.

Project Annexes become binding upon signature by authorized representatives of both Parties and do not require amendment of this Guide.

Annex B - Framework Guides and Prior Instruments

Where this Guide is executed within the scope of a broader institutional cooperation agreement or memorandum of understanding previously signed by the Lead Research Organization and the Partner (or their respective parent institutions), the terms of such prior instrument, where relevant, apply and are not superseded by this TCA. Any such prior agreements shall be listed and annexed hereupon execution.

Annex C - Regulatory and Compliance Policy Framework

Projects submitted to public Funding Bodies under this Guide are subject to the standard terms and conditions of the applicable funding body. The policies referenced in this Annex reflect internationally recognized good practice standards for research governance. They are provided as a reference framework. The Partner is expected to be aware of and act consistently with the spirit of these standards, and to comply with equivalent policies applicable in its own jurisdiction. References and links are provided for each area.

C.1 - Data Protection Policy

UKRI's Data Protection Policy applies to all personal data processed throughout its life cycle - from collection to destruction. It establishes principles that UKRI and its funded researchers must adhere to, covering roles and responsibilities, data subject rights (including rights of access), data sharing, and privacy-by-design approaches.

- Scope: All personal data processed in connection with publicly funded research and collaborative activities.
- Key obligation: Process personal data lawfully, fairly, and transparently; implement appropriate safeguards; respect data subject rights.
- Relevance to this Guide: Any personal data exchanged or processed in joint Projects must comply with the applicable data protection framework of each Party's jurisdiction (see Clause 9.4).

C.2 - GDPR and Research: An Overview for Researchers

This section summarises internationally recognized principles for data protection in research contexts, drawing on guidance from the UK Information Commissioner's Office (ICO) and equivalent international bodies. The principles are broadly applicable across jurisdictions.

- Key principles: lawfulness, fairness, transparency; purpose limitation; data minimization; accuracy; storage limitation; integrity and confidentiality.
- Research privileges: UK GDPR recognizes that research may require data to be processed and retained beyond its original purpose. Research is exempt from purpose and storage limitations provided appropriate safeguards are met.
- Anonymization: Properly anonymized data falls outside the scope of the UK GDPR. The ICO provides guidance on robust anonymization standards.
- Cross-border transfers: Requires an appropriate mechanism recognized under the applicable data protection law - such as Standard Contractual Clauses, adequacy decisions, or binding corporate rules.
- DPIAs: Required where processing is likely to result in high risk. The Lead Research Organization's DPO supports researchers in conducting DPIAs.

C.3 - Environmental Sustainability

Funding bodies and research institutions worldwide are increasingly committed to ensuring that the research and innovation they support contribute positively to environmental sustainability. Projects under this Guide are expected to consider their environmental impact in design, delivery, and dissemination.

- Relevance to this Guide: Projects involving high-performance computing (e.g., LLM inference, large-scale benchmarking) should consider energy efficiency. Approaches that reduce unnecessary computational cycles directly reduce energy consumption and environmental impact.

- The Partner is encouraged to report on environmental sustainability impacts as part of annual project reviews.

C.4 - Equality, Diversity and Inclusion (EDI)

Promoting equality, diversity, and inclusion (EDI) is a responsibility of all parties in a research collaboration. Funders and research institutions globally recognize that diverse teams produce more innovative and impactful research. This Guide encourages applications and nominations from a diverse range of researchers and supports inclusive working practices.

- Both Parties commit to applying EDI principles in the selection of individuals nominated under this Guide - including Visiting Researchers, Specialists, and Research & Innovation Associates.
- The Partner's diversity policies should be consistent with applicable equality and non-discrimination legislation in its jurisdiction, and with the spirit of internationally recognized EDI principles.

C.5 - Funding Assurance Program

Responsible stewardship of public and private research funds is a shared obligation. Funding bodies worldwide implement assurance mechanisms - including post-award audits, spot checks, and outcome monitoring - to ensure that funds are properly safeguarded and used for intended purposes.

- Both Parties agree to maintain accurate financial records for all project-related expenditure and to cooperate fully with any audit or assurance activity initiated by the applicable Funding Body.
- The Partner, as a non-grant-receiving organization, must be able to document the value and nature of its declared direct and indirect contributions upon request by the Funding Body or the Lead Research Organization.

C.6 - Information Management Policy

Effective information management supports core research functions, ensures compliance with legal obligations, and contributes to good organizational governance. UKRI's Information Management Policy governs how research information, outputs, and records are created, stored, accessed, shared, and destroyed.

- Research outputs (publications, datasets, code, reports) arising from funded Projects must be managed in accordance with the Funding Body's open-access and open-data policies and applicable institutional requirements.
- Both Parties should agree on a data management plan at the start of each Project, covering where data is stored, who can access it, and how it will be disposed of at the end of the Project.

C.7 - Principles of Public Life (Nolan Principles)

Good governance in research is guided by internationally recognized principles of public and professional life: selflessness, integrity, objectivity, accountability, openness, honesty, and leadership. These principles - codified in frameworks such as the UK's Nolan Committee (UK) principles and equivalent international standards - govern the conduct of all individuals involved in publicly funded research.

- All researchers, partner representatives, and personnel acting under this Guide are expected to act consistently with these principles and with equivalent standards applicable in their own jurisdictions.
- Particular attention is drawn to integrity and openness: research results must not be suppressed or selectively reported to serve commercial interests at the expense of scientific accuracy.

C.8 - Modern Slavery and Human Trafficking

Preventing modern slavery and human trafficking is a global responsibility. These are serious, often hidden crimes with devastating consequences for victims. Both Parties commit to taking active steps to prevent modern slavery and human trafficking in their operations and supply chains, in accordance with applicable law.

- Both Parties confirm that they have taken reasonable steps to ensure that modern slavery and human trafficking are not taking place in their own operations or supply chains.
- Any credible concern regarding modern slavery in connection with activities under this Guide must be reported immediately to the relevant institutional authority and, if applicable, to the Funding Body or applicable regulator.

C.9 - Policy on Use of Animals in Research

Many funding bodies maintain policies on the ethical use of animals in research. The internationally recognized 3Rs framework - Replace, Reduce, Refine - provides guidance for the ethical treatment of animal subjects, and is embedded in research ethics frameworks across the world.

Relevance to this Guide: Projects under this Guide focus on software verification, formal methods, and AI and are not expected to involve animal subjects. This policy is included for completeness and in the event that the scope of cooperation is expanded.

How to use these Appendices

These appendices provide a concise reference to internationally recognized regulatory and compliance standards applicable to publicly funded research collaborations. They are not exhaustive. Where a specific Project Annex requires a more detailed compliance plan (e.g., a Data Management Plan, DPIA, or EDI monitoring framework), those documents shall be prepared at the project level and attached to the relevant Annex.